

ECE 3210 quiz05

Name: _____ SOLUTIONS

1. (10 points) Evaluate the integral

$$I = \int_{-\infty}^{\infty} \text{sinc}(t) dt.$$

Solution: Here we recognize that this integral is a product of two signals $f_1(t) = 1$ and $f_2(t) = \text{sinc}(t)$. We can apply the generalized Parseval's theorem to evaluate this integral. We have

$$I = \int_{-\infty}^{\infty} f_1(t) f_2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_1(\omega) F_2(\omega) d\omega$$

We know that $F_1(j\omega) = 2\pi\delta(\omega)$ and $F_2(j\omega) = \pi \text{rect}\left(\frac{\omega}{2}\right)$. Substituting these into the integral gives

$$I = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega) \pi \text{rect}\left(\frac{\omega}{2}\right) d\omega$$

We see that the 2π terms cancel and we can pull out the π term. This gives an integral

$$I = \pi \int_{-\infty}^{\infty} \delta(\omega) \text{rect}\left(\frac{\omega}{2}\right) d\omega$$

This is just a sifting problem, which gives $I = \pi$.

We can also look at this problem from a different perspective. We can recognize that

$$\int_{-\infty}^{\infty} \text{sinc}(t) dt = \int_{-\infty}^{\infty} \text{sinc}(t) e^{-j\omega t} dt \Big|_{\omega=0}$$

This tells that we can simply take the Fourier transform of the $\text{sinc}(t)$ function and evaluate it at $\omega = 0$. We know that the Fourier transform of $\text{sinc}(t)$ is $\pi \text{rect}\left(\frac{\omega}{2}\right)$. Evaluating this at $\omega = 0$ gives $I = \pi$.